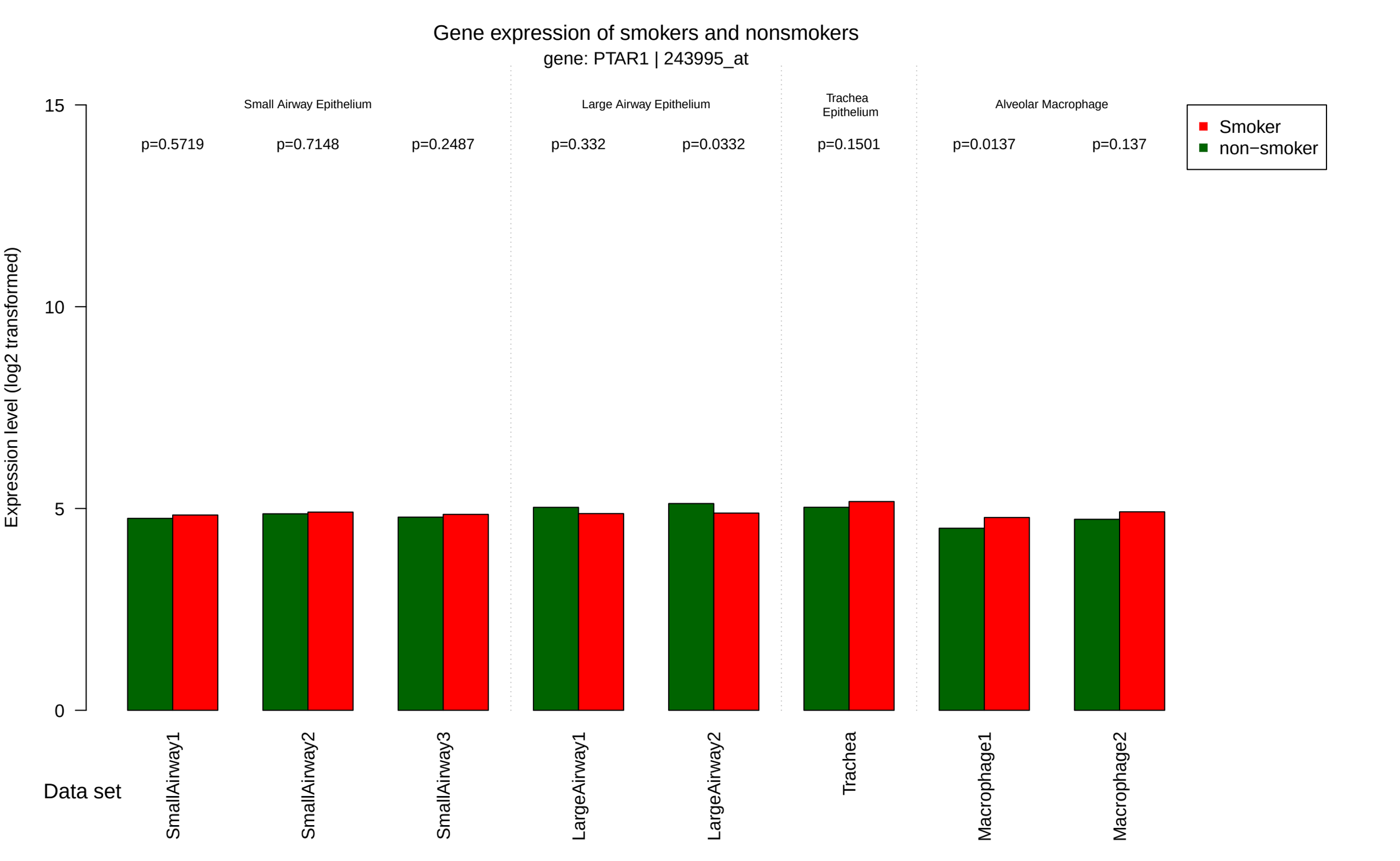
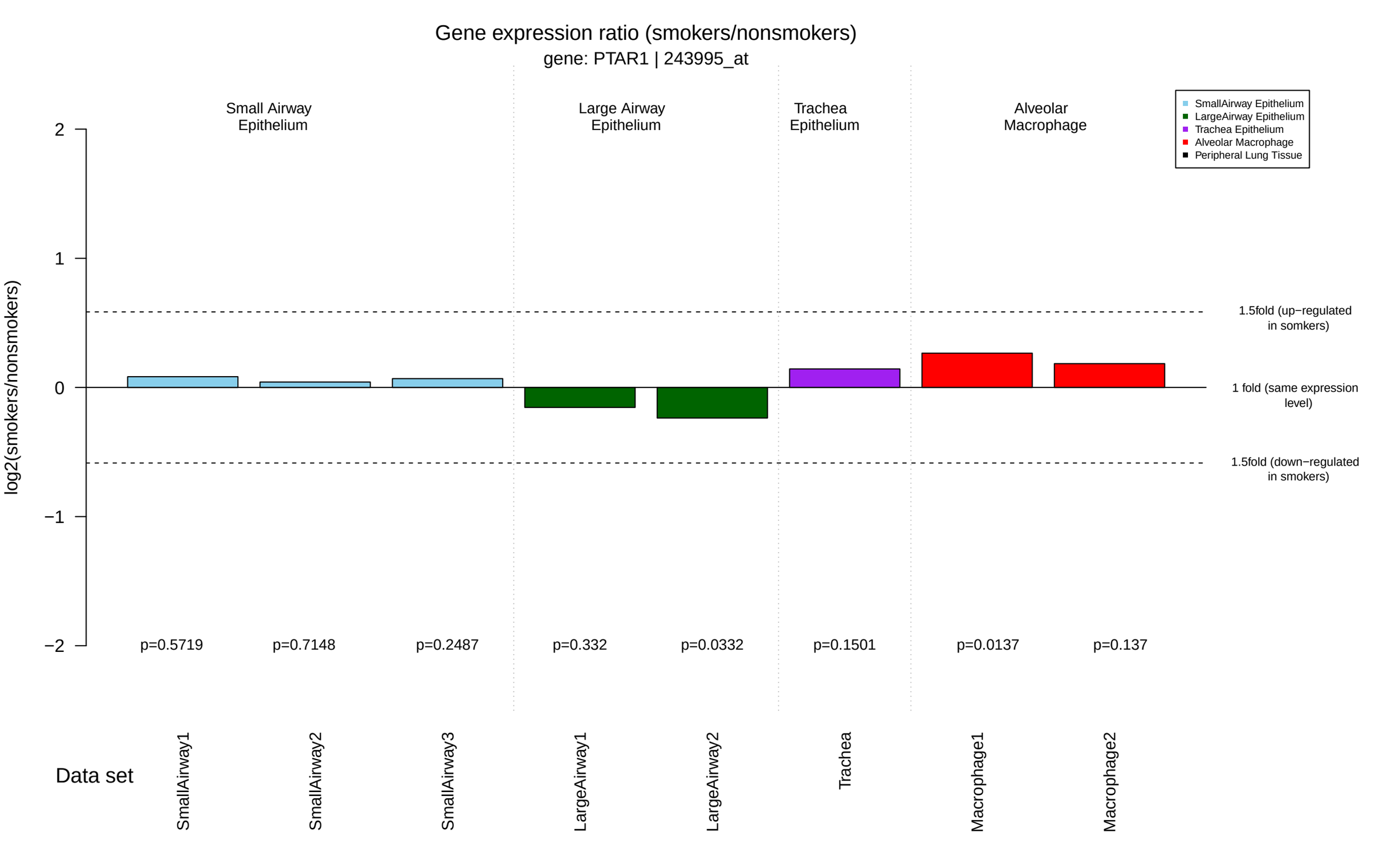
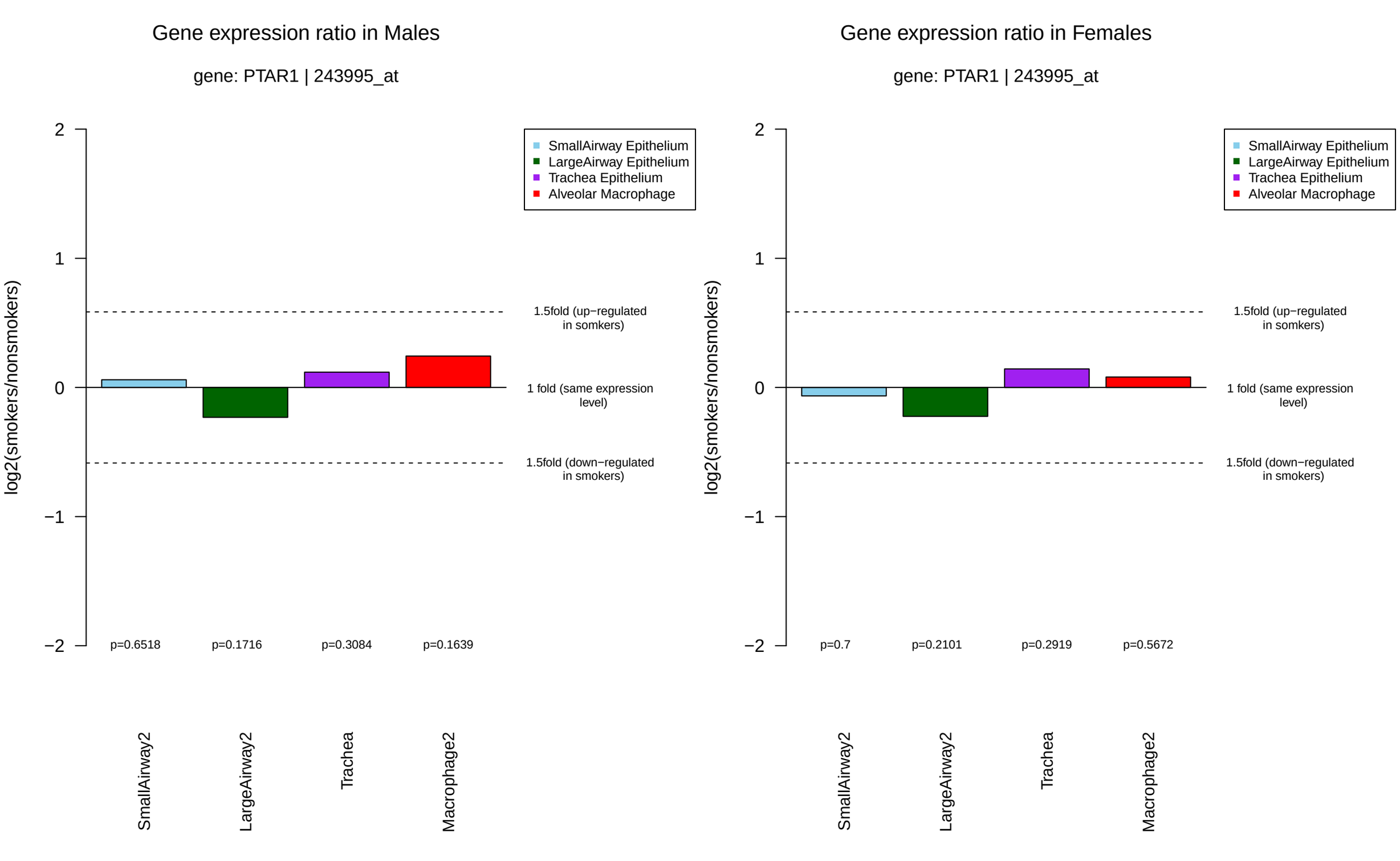
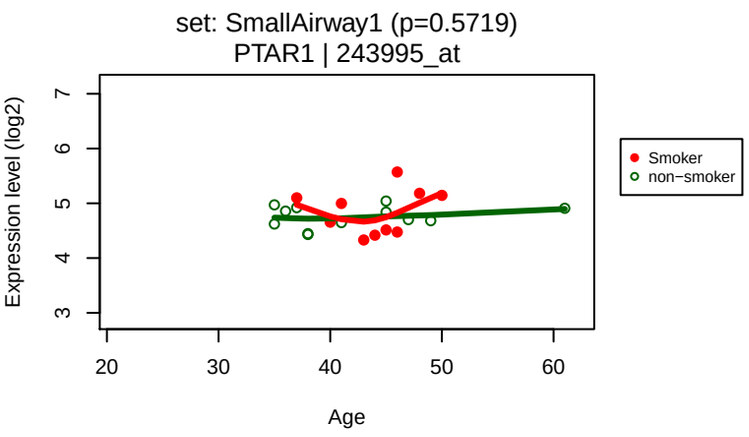
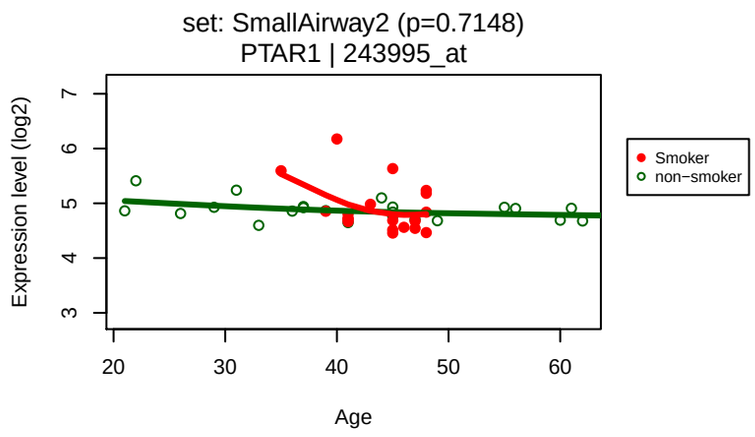
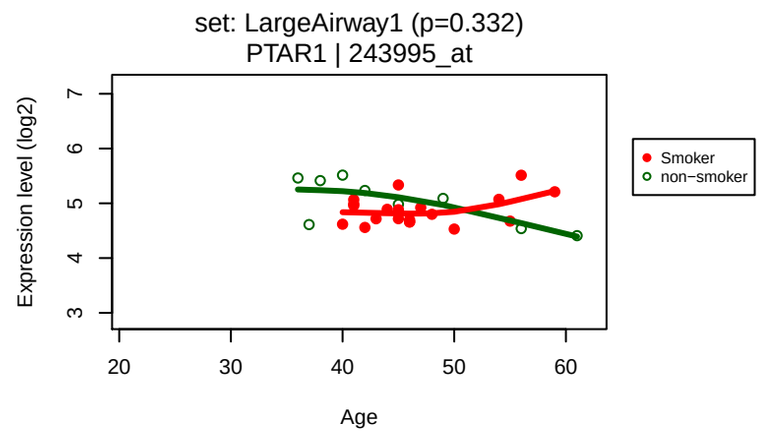

Gene expression of smokers and nonsmokers
gene: PTAR1 | 243995_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5719
p=0.7148
p=0.2487
p=0.332
p=0.0332
p=0.1501
p=0.0137
p=0.137
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

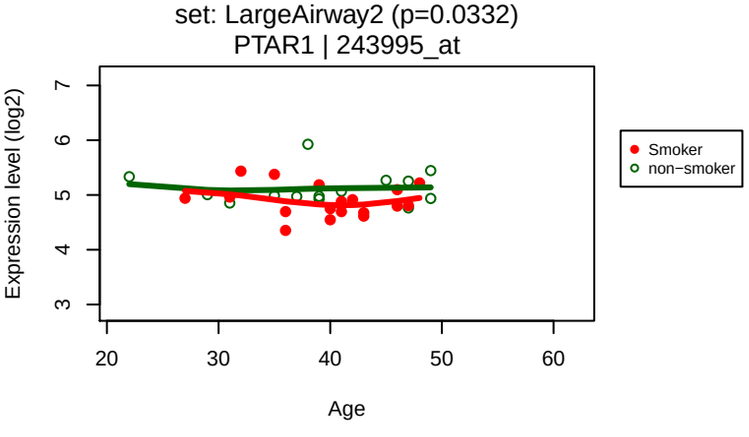

Gene expression ratio (smokers/nonsmokers)
gene: PTAR1 | 243995_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5719
p=0.7148
p=0.2487
p=0.332
p=0.0332
p=0.1501
p=0.0137
p=0.137
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

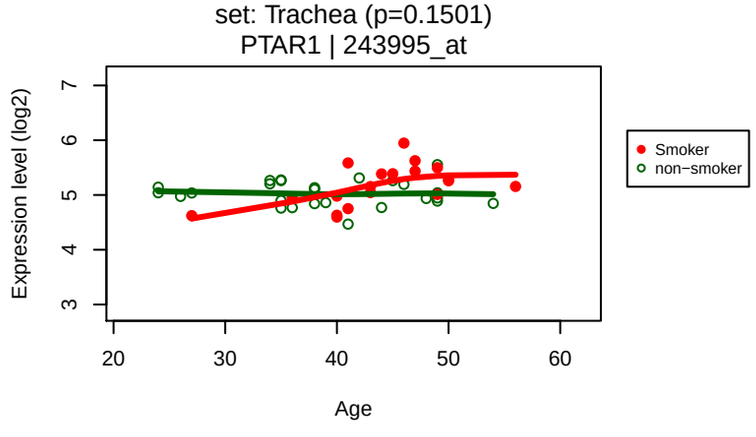

Gene expression ratio in Males
gene: PTAR1 | 243995_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6518
p=0.1716
p=0.3084
p=0.1639
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTAR1 | 243995_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7
p=0.2101
p=0.2919
p=0.5672
SmallAirway2
LargeAirway2
Trachea
Macrophage2

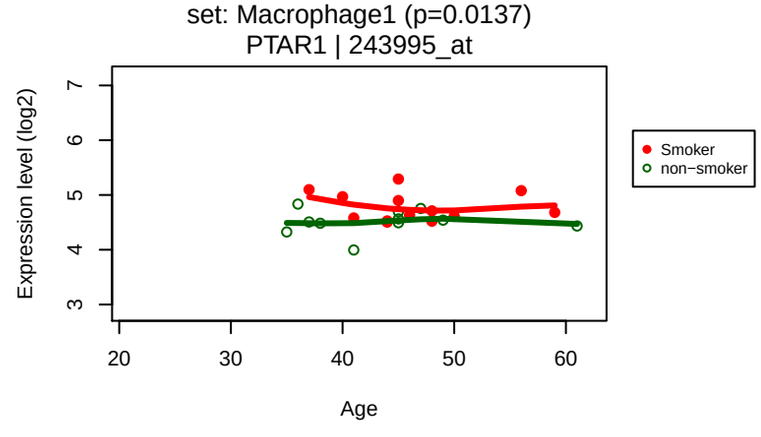

set: SmallAirway1 (p=0.5719)
PTAR1 | 243995_at
Expression level (log2)
Age
Smoker
non−smoker

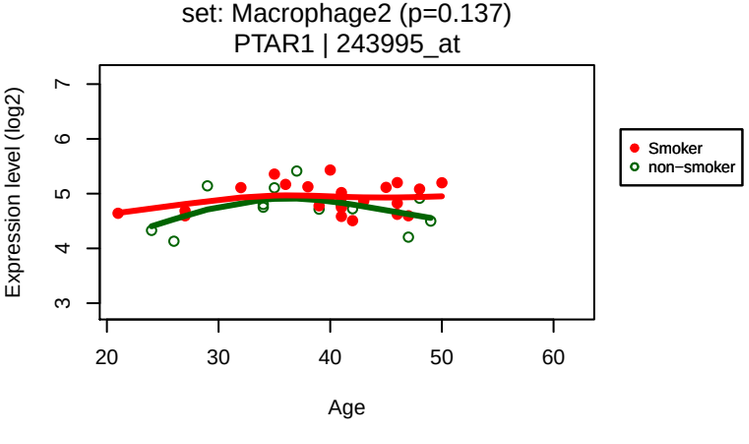

set: SmallAirway2 (p=0.7148)
PTAR1 | 243995_at
Expression level (log2)
Age
Smoker
non−smoker

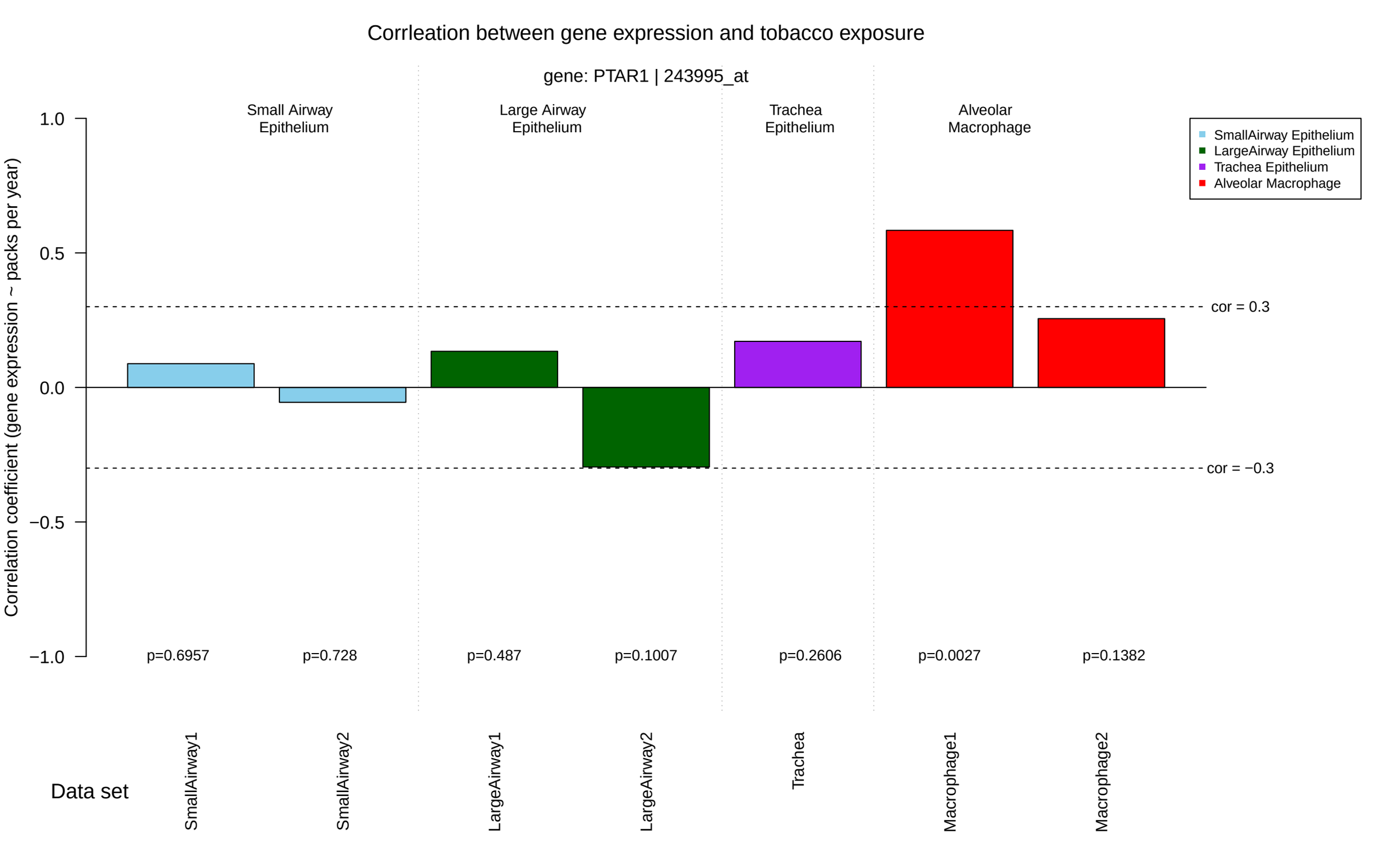

set: LargeAirway1 (p=0.332)
PTAR1 | 243995_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0332)
PTAR1 | 243995_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1501)
PTAR1 | 243995_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0137)
PTAR1 | 243995_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.137)
PTAR1 | 243995_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTAR1 | 243995_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6957
p=0.728
p=0.487
p=0.1007
p=0.2606
p=0.0027
p=0.1382
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2